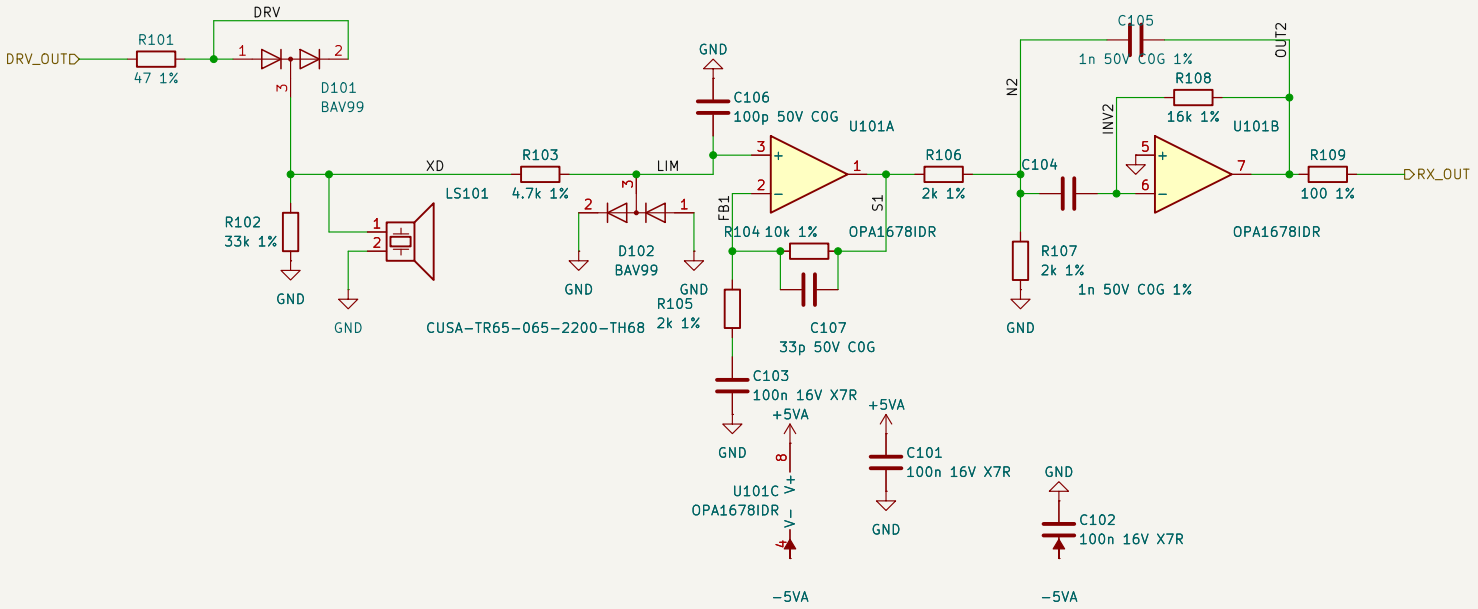
Transceiver channel (one of 64). TX: UCC27524 half -> 47 R -> BAV99 crossed diodes -> element (0-15 V square).
33k bleed resets the element after TX. RX: 4.7k + BAV99 clamp (+-0.7 V) + 100 pF (340 kHz RF filter).
OPA1678 stage 1: gain 6, AC-coupled at 800 Hz. Stage 2: MFB band-pass f0 39.8 kHz, Q 2, gain -4. Then 100 R to the ADC.
Element -> ADC +27 dB, pass band 31-51 kHz (sim/reports/rx_chain.md). Band-pass parts 1 % C0G for phase match.



Hybrid 8x8 ultrasonic AESA
Sheet: /Transceivers/CH01/
File: channel.kicad_sch
Title: Transceiver channel (x64): TX, T/R switch, RX amplifier
Size: A4 | Date: 2026-10-20 | Rev: 2
KiCad E.D.A. 10.0.6 | Id: 33/115